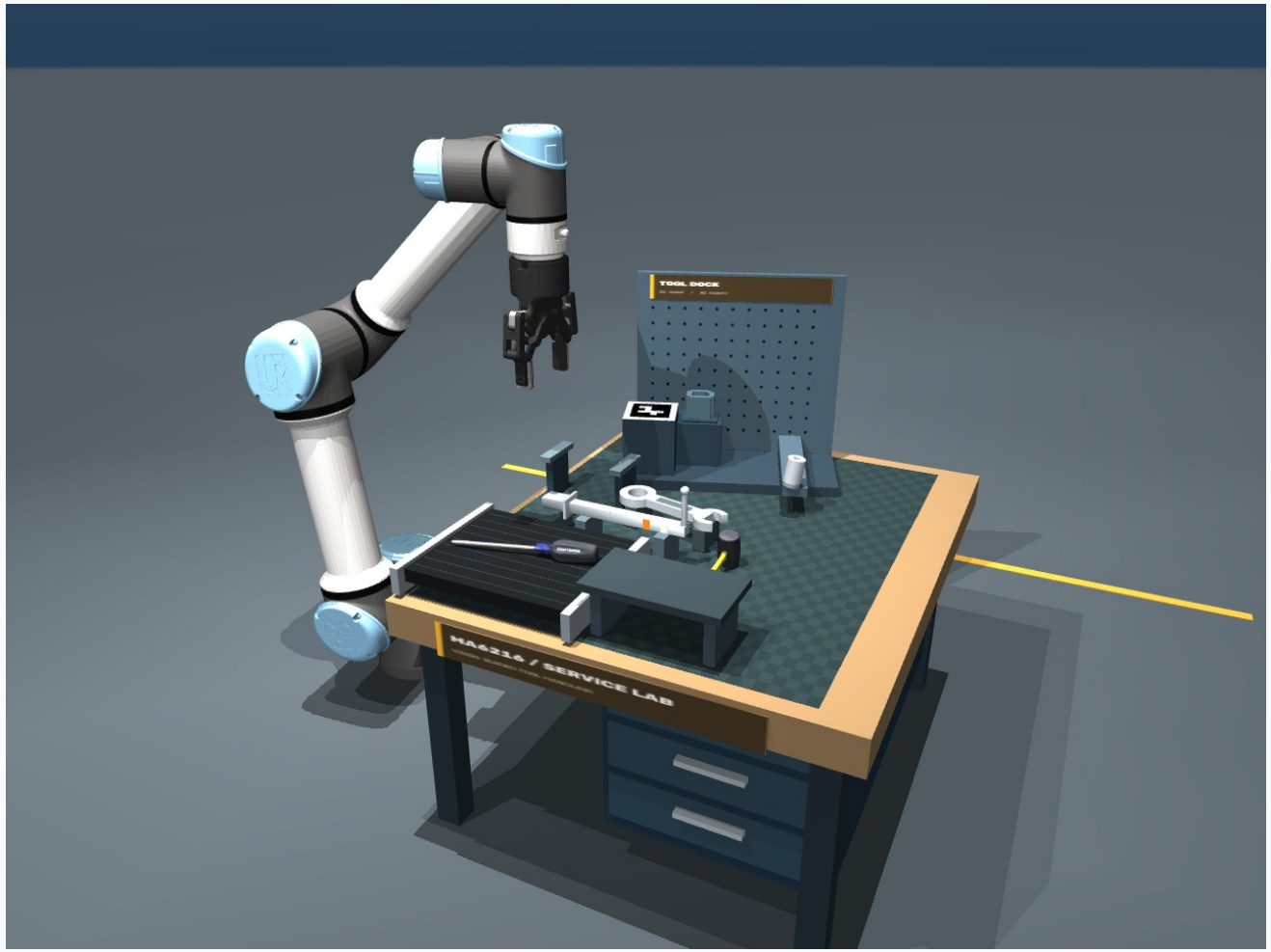
Vision-Guided Tool Manipulation



Calibrate

Localize

Estimate 6D pose

Manipulate

Evaluate

The task requires orientation, not only position

1 / Return	2 / Insert	3 / Assemble	4 / Regrasp + hang
Hammer tray pose	Moving screwdriver Tilted socket	Independent rack Relocalize after seating	Actual finger release Hole-over-pin placement

Final acceptance: geometry + support contact + release + stability

System architecture and evaluation boundary

- RGB-D + estimated calibration → foreground mask
- Reference reconstruction + FoundationPose → camera-frame 6D pose
- Camera-to-robot transform → grasp / insertion target
- IK + collision checks + joint control → physical manipulation
- Reobserve → bounded pose correction; score with independent truth

Technical selection: match outputs to the task

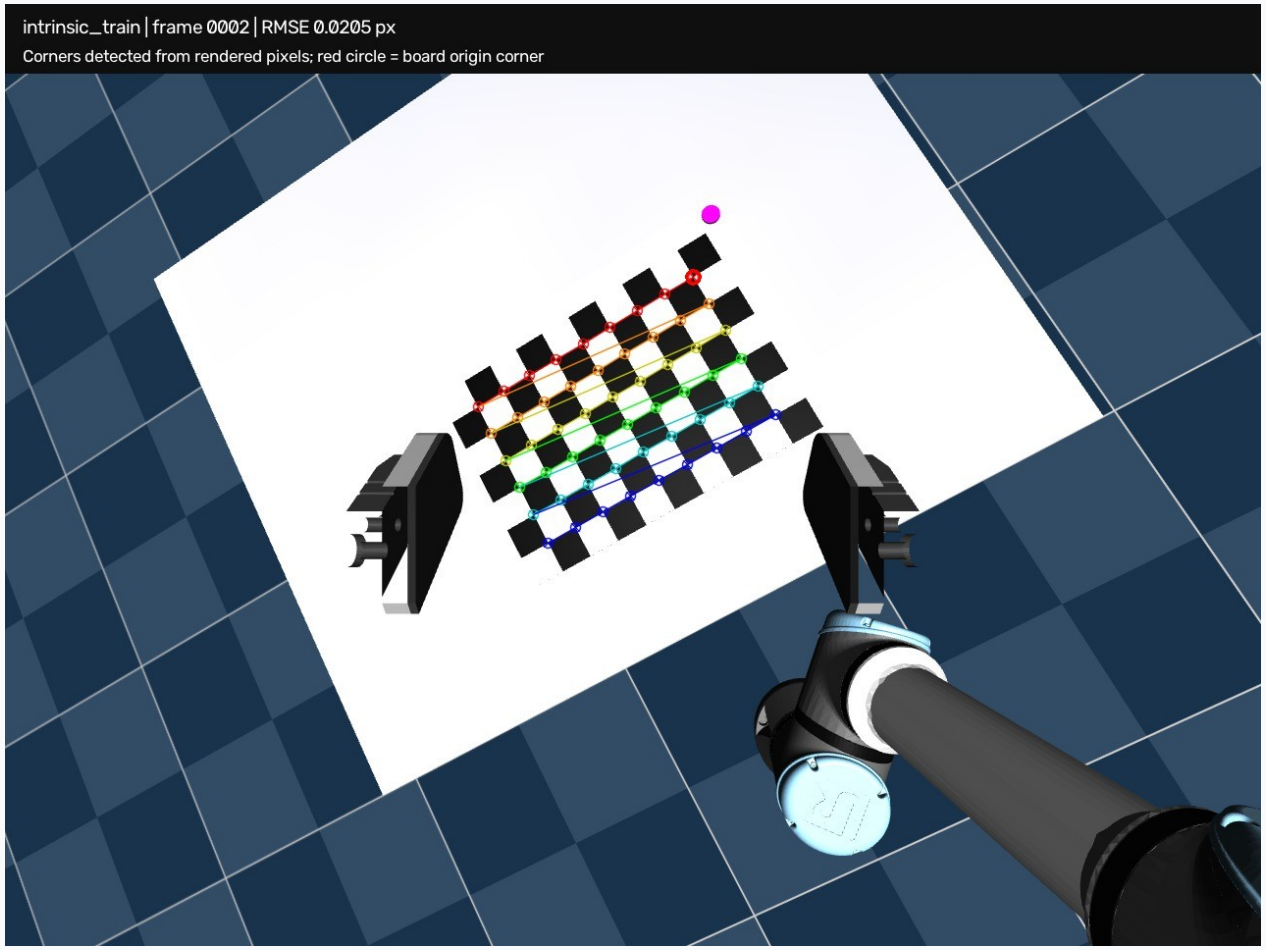
Candidate	Task fit	Project decision
FoundationPose	Novel-object 6D pose + tracking	Selected; RGB-D reference route
MegaPose	CAD + object region → 6D pose	Reviewed; not experimentally compared
GraspNet	General grasp candidates	Reviewed; future pickup alternative
Depth / color geometry	Foreground and coarse geometry	Depth mask selected; not full pose

Wen et al., CVPR 2024; Labbé et al., arXiv:2212.06870; Fang et al., CVPR 2020

Why MuJoCo and a Python integration?

- Reuse the validated UR5e / Robotiq workcell and contact acceptance
- Render RGB-D and replay saved physical states
- Keep camera, vision and controller interfaces inspectable
- Alternatives reviewed; no simulator accuracy or speed ranking
- Limit: trajectory templates and checked IK, not global planning

Task 2 / Image-based camera intrinsics



Wrist: 24 train / 8 validation

0.02177 px validation RMSE

113 acquired images retained

66 accepted across all splits

Zhang, IEEE TPAMI 2000; docs/results/wrist-calibration-v1

Task 3 / Estimate the wrist-to-gripper transform

$$G_i X C_i = Y \quad \rightarrow \quad AX = XB$$

- 24 valid training views; 10 independent hand-eye validation views
- Estimated hand-eye error: 0.852 mm / 0.0227°
- Frozen-chain validation: 0.05497 px
- Runtime: world_T_gripper(FK) × gripper_T_camera(estimated)

Task 4 / Independent localization evaluation

Condition	Mean visible-mask IoU	Scene-bootstrap 95% CI
Nominal	0.887	0.854–0.923
Position + orientation	0.917	0.883–0.953
Partial occlusion	0.810	0.808–0.812
Depth noise	0.810	0.785–0.838

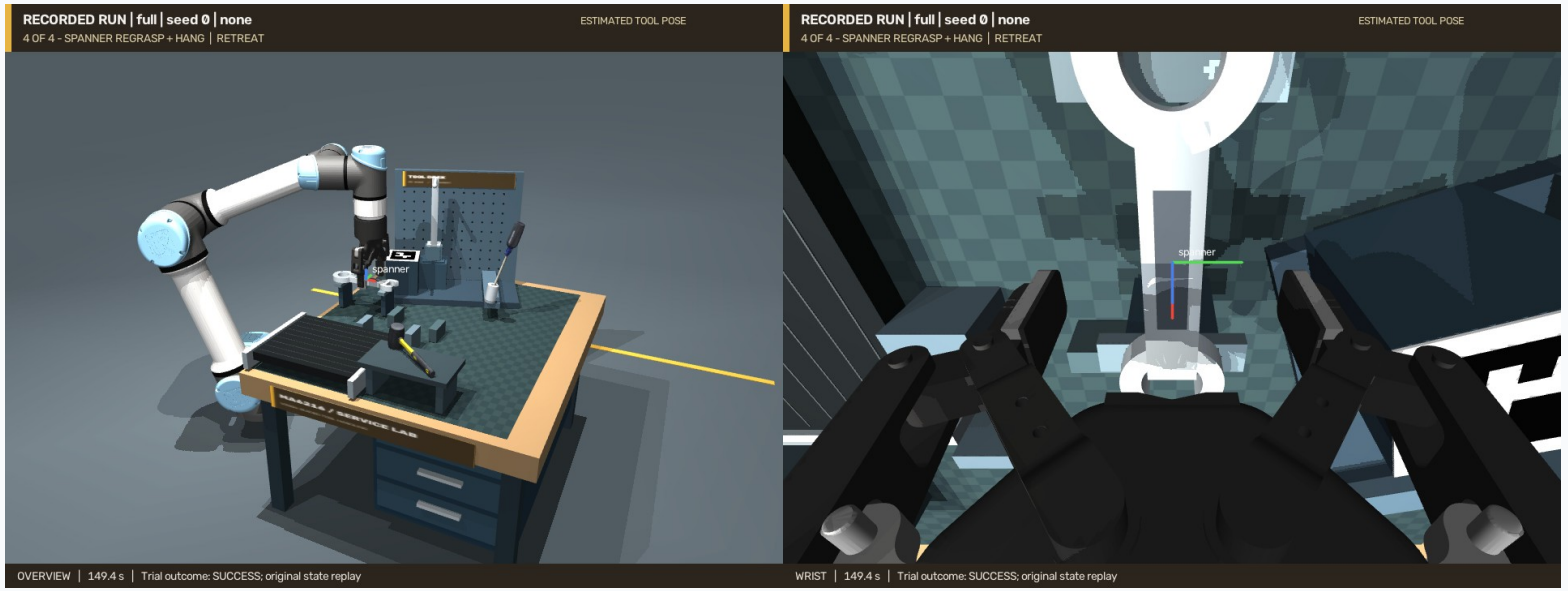
160/160 positive queries passed IoU ≥ 0.5; 0/40 absent-target false positives.

Each object / condition: n = 10. Background-mismatch failures archived separately.

Task 5 / What “model-free” means here

- 24 reference RGB-D views + masks + scan calibration
- Official Neural Object Field → reconstructed object surface
- Official FoundationPose register / track → camera_T_object
- Estimated task-camera transforms → robot-frame target
- Ground-truth translation / rotation / ADD / ADD-S for evaluation

Task 6 / Physical release and regrasp matter



Plan → grip → transfer

Observe → release → regrasp

Align → insert / hang

Release → verify support

Bonus / Three levels of visual feedback

Mode	Visual correction during manipulation
Open loop	Initial acquisition / feed prediction only
Staged	Reobserve at selected task transitions
Iterative	Staged + repeated bounded 6D alignment

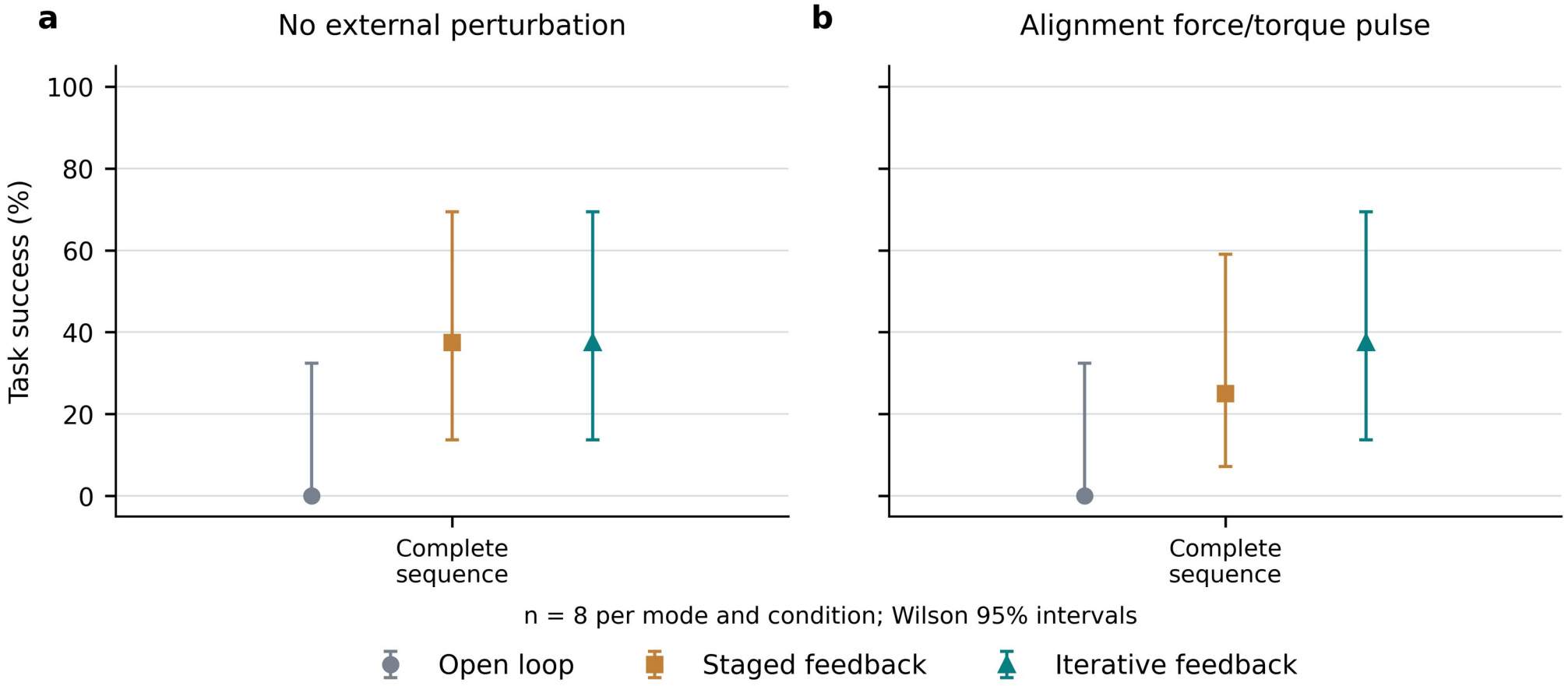
Default correction cap: 4 mm / 2° per step

Convergence: two observations within 1.5 mm / 1.5°

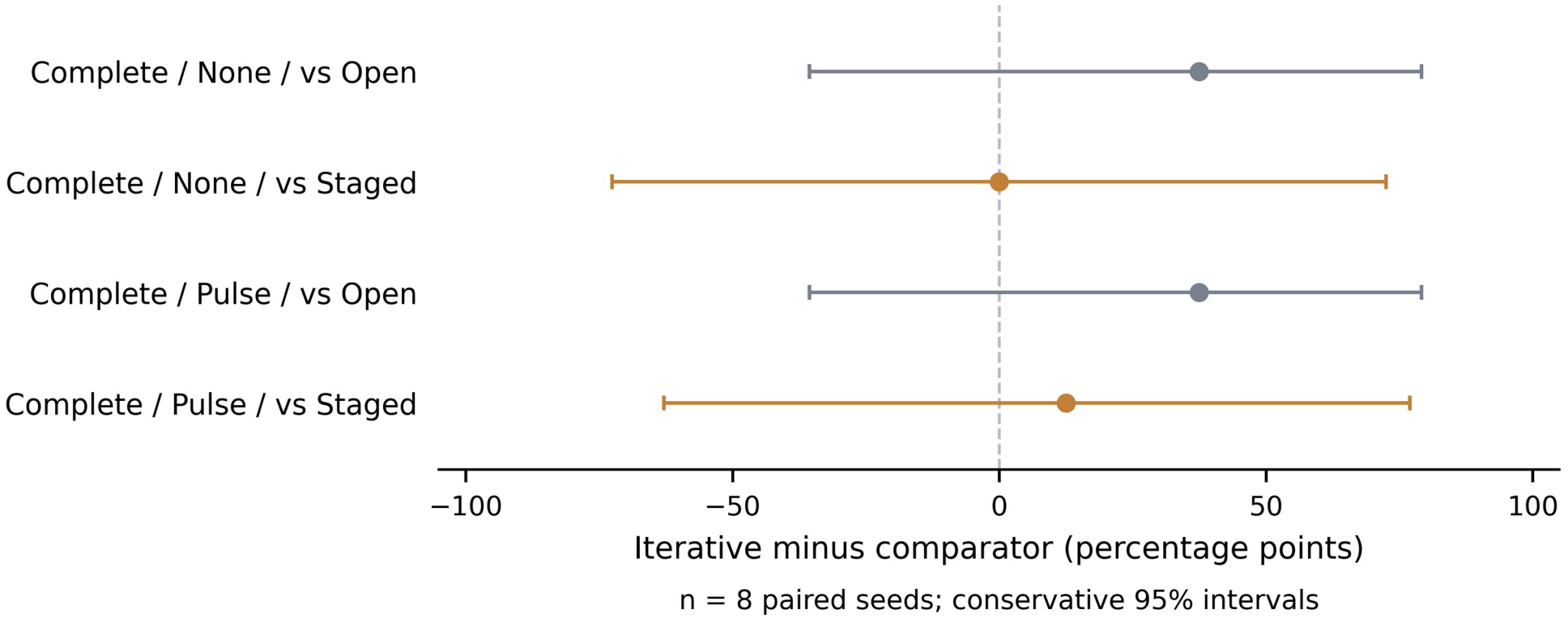
Experimental design: 48 paired attempts

- 3 modes × 2 conditions × 8 seeds = 48 attempts
- Estimated wrist calibration; fixed control and acceptance thresholds
- Same initial states paired by seed; disturbance delivery logged
- Wilson 95% CI; bootstrap trial means; paired exact tests + Holm
- Seeds 400–407 reused; separate from historical ideal-wrist results

Whole-sequence success with uncertainty



Paired differences bound the conclusion



Accuracy and cost are separate measurements

Mode (no pulse)	Pose / mm	Pose / deg	TCP RMSE / mm	Pose calls / wall s
Open loop	2.60 (n=8)	1.44 (n=8)	5.33 (n=8)	0.04 (n=8)
Staged	2.60 (n=8)	1.44 (n=8)	5.24 (n=8)	0.03 (n=8)
Iterative	2.60 (n=8)	1.44 (n=8)	5.23 (n=8)	0.04 (n=8)

Physics: 500 Hz ≠ visual feedback frequency

True alignment errors and visual residuals are recorded separately.

Full conditions, confidence intervals and sample counts are in the accompanying report

Failure analysis: where the complete system stops

Last recorded stage	Failed attempts
CLOSE_GRIPPER	24
PICK_APPROACH_SETTLE	6
RETRACT_DRIVER_SETTLE	6
REORIENT_AND_APPROACH_RACK	1

Limits and the next engineering priorities

- Known stations and paired empty backgrounds; no arbitrary clutter claim
- Ideal reference scans, FK and rigid mounts; simulation only
- Small reused-seed sample; no architectural superiority assumption
- Priority: pickup reliability → pose bias / occlusion → feedback timing
- External pose baselines and real hardware remain future work

Deliverables and course coverage

- Task 1: literature, alternatives and selection rationale
- Tasks 2–3: image-estimated intrinsics and hand-eye calibration
- Tasks 4–5: localization audit and 6D pose accuracy
- Task 6 + Bonus: complete manipulation and paired feedback evaluation
- Editable slides, English notes, data, code, videos and limitations

References and reproducibility

- Wen et al. — FoundationPose, CVPR 2024
- Labbé et al. — MegaPose, arXiv:2212.06870, 2022
- Fang et al. — GraspNet-1Billion, CVPR 2020
- Zhang — Camera calibration, IEEE TPAMI, 2000
- Chaumette & Hutchinson — Visual servo control I, IEEE RAM, 2006
- Todorov et al. — MuJoCo, IROS 2012

Full bibliographic details and primary links: docs/LITERATURE_AND_SELECTION.md